CATHOLIC JUNIOR COLLEGE
JC2 Preliminary Examinations
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME				
CLASS	2T			

INDEX NUMBER				

BIOLOGY 9648/02

Paper 2 Core Paper

28 August 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units

At the end of the examination, fasten all work securely together.
The number of marks is given in brackets [ ] at the end of each question or part of the question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	

This document consists of **21** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1(a) Explain the need for reduction division in meiosis prior to fertilisation in sexual reproduction.

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Fig. 1.1 below shows 8 different micrographs taken at the different timings during the meiosis of a cell.

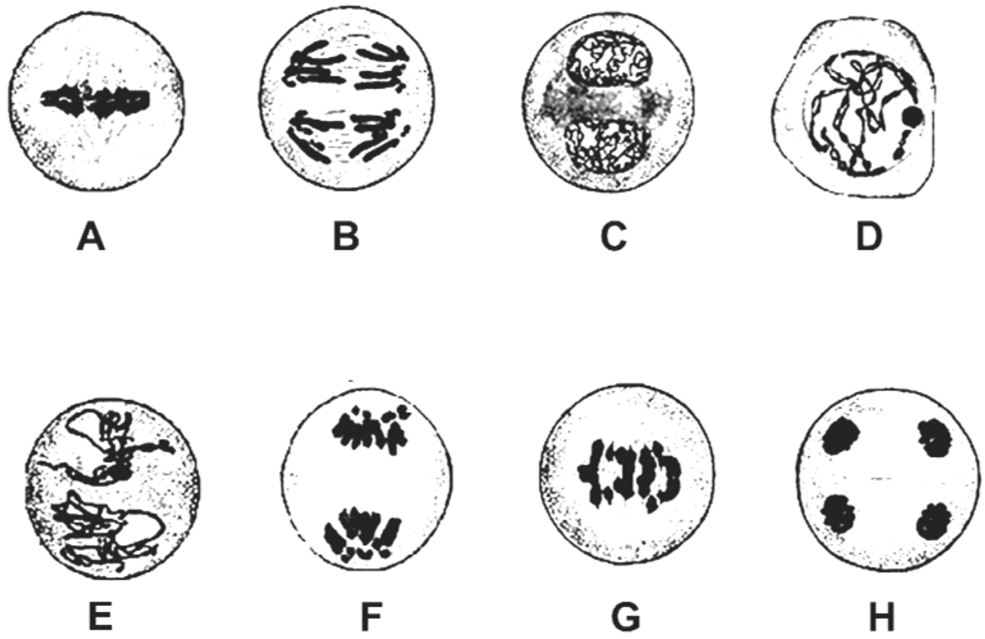

Fig. 1.1

(b) Arrange the 8 stages (**A** - **H**) in chronological order.

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(c) With reference to Fig. 1.1,

(i) identify **one** stage (**A** - **H**) that will result in genetic variation.

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(ii) explain how this stage results in genetic variation.

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Bone marrow contains stem cells that divide by mitosis to form blood cells. Each time a stem cell divides it forms a replacement stem cell and a cell that develops into a blood cell.

Fig. 1.2 shows changes in the mass of DNA in a human stem cell from the bone marrow during three cell cycles.

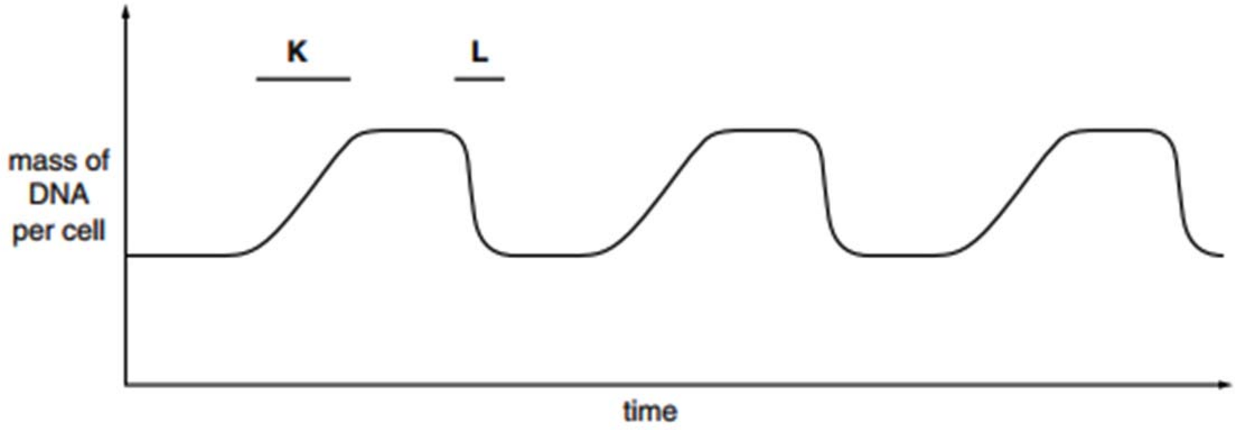

Fig. 1.2

(d) With reference to Fig. 1.2, state

(i) what happens to bring about the changes in the mass of DNA per cell at **K** and at **L**.

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(ii) how many blood cells are formed from the stem cell in the time shown.

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(iii) what happens to the number of chromosomes in the stem cell.

....[1]

[Total: 10]

2 Phosphate ions have a number of uses in organisms. These include:

- involvement in cell signalling responses
- involvement in energy transfer processes
- component of phospholipids
- component of nucleotides

Phosphatase enzymes remove phosphate groups from organic compounds, while kinase enzymes add phosphate groups.

(a) Read the following passage:

Some hormones circulating in the blood are able to trigger transcription within a cell, even though they are unable to enter the cell. Phosphatases and kinases then take part in cell activities that eventually result in genes switching on and transcription beginning.

(i) Suggest why the hormones, referred to in the passage, are unable to enter the cell.

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(ii) Use the information in the passage to outline the process of cell signalling.

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(b) The activity of a phosphatase enzyme was measured at different values of pH by using nine different buffer solutions. The temperature was kept constant at 30 ^{o}C.

The results are shown in Fig. 2.1.

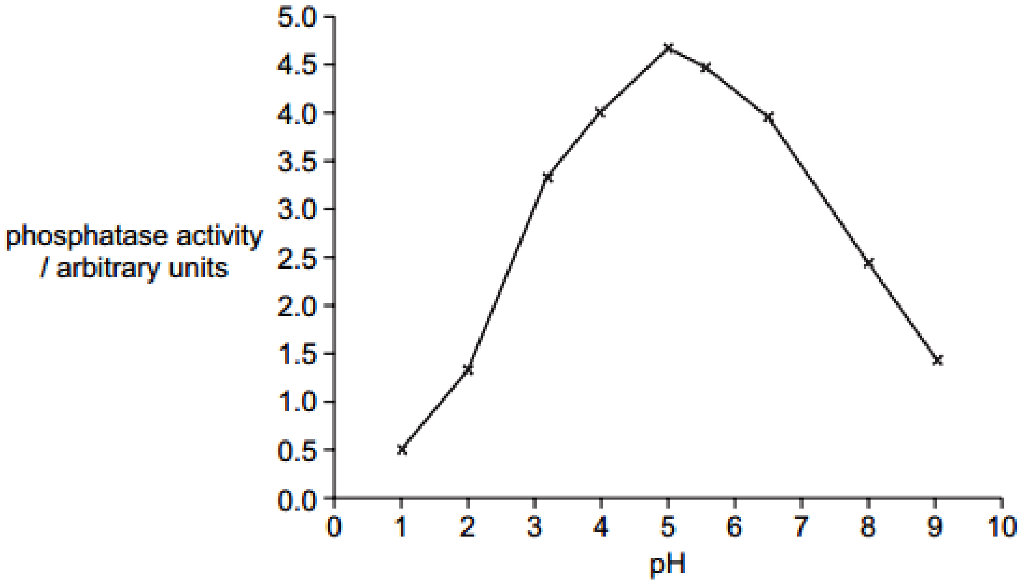

Fig. 2.1

(i) With reference to Fig. 2.1, describe the effect of pH on the activity of phosphatase.

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(ii) Explain why the activity of phosphatase at pH 1 is very low.

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....[3]

(c) Phosphatases are being used commercially to produce dephosphorylated DNA. This involves removal of the two terminal phosphate groups using phosphatases.

(i) In the box below, draw a labelled diagram of a dephosphorylated DNA nucleotide containing guanine base. [1]

(ii) Compare the structure of an ATP molecule with the DNA nucleotide drawn in **(c)(i)**.

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[Total: 15]

3 CFTR gene is controlled by a promoter that has a variable proximal region. Detailed investigation reveals that gene expression is dependent on the conservation of a particular sequence within the proximal region. Table 3 shows the results of the investigation.

Table 3

Individual	Proximal region of the promoter sequence	Binding of RNA polymerase II	Expression of CFTR
1	5' – AATTGGAAGCAAAT – 3'	Able to bind	Basal level
2	5' – CCTTGGAAGCAAAT – 3'	Able to bind	Basal level
3	5' – TTTTGGAAGCAAAT – 3'	Able to bind	Basal level
4	5' – AATTAAGGTCAAAT – 3'	Able to bind	Basal level
5	5' – AATTAAGGTCATTT – 3'	Unable to bind	Absent
6	5' – AATTAAGGTTGAAC – 3'	Unable to bind	Absent
7	5' – AATTGGAAGCAAAC – 3'	Unable to bind	Absent

(a) With reference to Table 3;

(i) State the conserved sequence that must present for the CFTR gene to be expressed.

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(ii) Explain why gene expression was absent in individual **5**.

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Gene mutation responsible for cystic fibrosis can be classified into 6 different classes.

Class I and **II** CFTR mutations are typically characterised by a reduction in the quantity of functional CFTR protein.

- **Class I** mutations can result from nonsense and frame-shift mutations, as well as mRNA splicing defects.
- **Class II** mutations are caused by folding or maturation defects, which can result in premature CFTR degradation.

Class III and **IV** mutations, however, are typified by aberrant channel function rather than reduced quantities of CFTR.

- CFTR channels with **Class III** mutations result in limited channel gating that arises from ineffectual binding of nucleotide.
- CFTR channels with **Class IV** mutations are able to open and close; however, chloride and bicarbonate ions are unable to freely pass through the channel due to conductance defects.

While **Class V** mutations can lead to the production of normal CFTR, a limitation of transcriptional regulation results in a reduced quantity of the protein being produced.

Finally, the relatively novel **Class VI** mutations are characterised by high turnover of CFTR at the cell surface.

(b) Explain how **Class I** mutations result in a reduction in the quantity of functional CFTR protein.

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(c) Suggest how **Class V** mutations result in a reduced quantity of protein formed.

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(d) Explain why a high turnover of CFTR at the cell surface results in cystic fibrosis.

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[Total: 9]

4 Fig. 4.1 shows how Histone 3 (H3) in eukaryotes may be modified for gene expression. Phosphorylation of H3 at position S10 results in rapid acetylation, which in turn leads to increased gene expression.

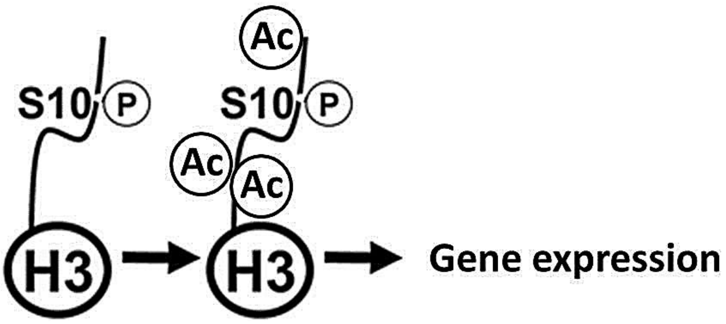

Fig. 4.1

(a) Suggest how the phosphorylation of H3 would result in the rapid acetylation of the protein.

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(b) Explain how the acetylation of H3 leads to increased gene expression.

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[Total: 6]

5 MRSA is a variety of *Staphylococcus aureus*. It is difficult to treat infections caused by this bacterium because it is resistant to methicillin and to some other antibiotics. As a result, some patients who are already very ill may die if they become infected with MRSA.

(a) The antibiotic methicillin inhibits the enzyme transpeptidase. This enzyme is used by some bacteria to join monomers together during cell wall formation. Methicillin has a similar structure to these monomers.

Use this information to explain how methicillin inhibits the enzyme transpeptidase.

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(b) Fig. 5.1 shows the number of deaths in England and Wales between 1994 and 2008 caused by MRSA.

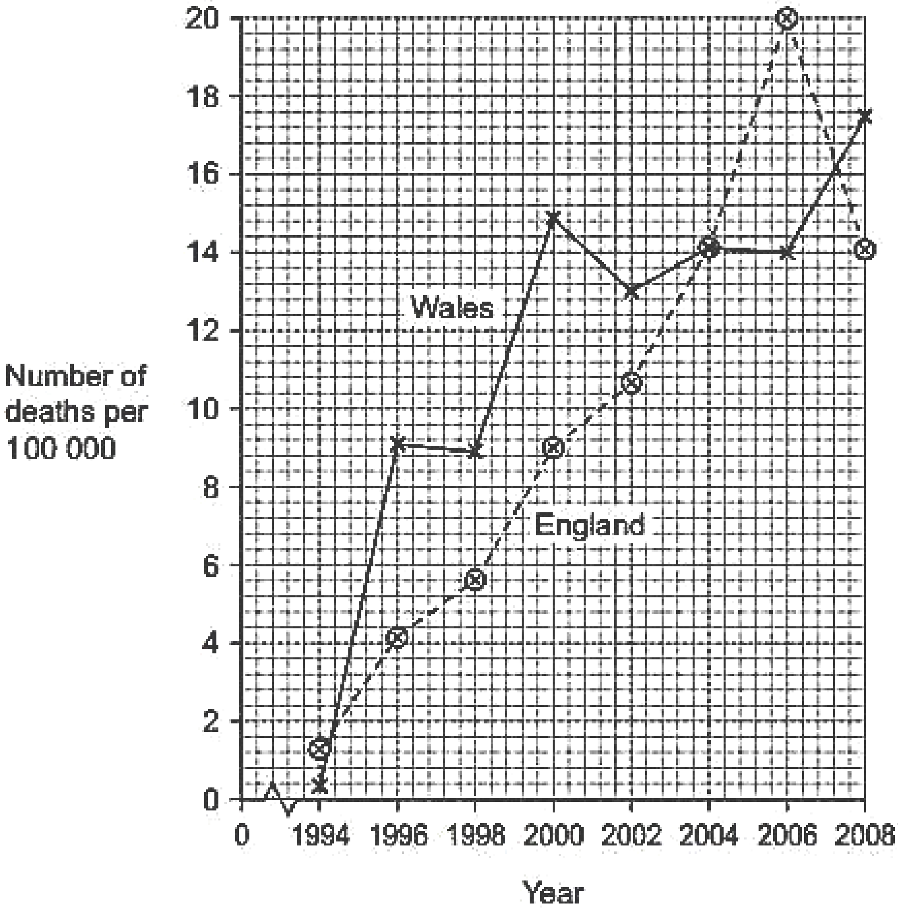

Fig. 5.1

(i) Describe the change in the number of deaths caused by MRSA in England in the period shown in the graph.

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(ii) Calculate the percentage increase in the number of deaths caused by MRSA in Wales from 1996 to 2006. Show your working.

Answer ……………………… [2]

(c) Describe how gene transmission and selection have increased the difficulty of treating bacterial infections with antibiotics.

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[Total: 10]

6 In mice, fur colour is controlled by a gene with multiple alleles. These alleles are listed below in no particular order.

yellow = C^y agouti = C^a black = C^b

(a) Suggest explanations for the results of the following crosses between mice.

(i) Mice with agouti fur crossed with mice with black fur may produce all agouti offspring or some agouti and some black offspring.

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(ii) Crosses between heterozygous parents with the genotype C^y C^b always produce a ratio of two yellow mice to one black mouse.

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(b) The budgerigar, *Melopsittacus undulatus*, is a small type of parrot that is native to Australia.

Fig. 6.1 shows a budgerigar.

Fig. 6.1

A budgerigar can have blue, green, yellow or white feathers.

Two genes, **A/a** and **D/d**, are involved in the inheritance of feather colour in budgerigars.

- A bird which has at least one dominant allele **A** but is homozygous for **d** has blue feathers.
- A bird which has at least one dominant allele **D** but is homozygous for **a** has yellow feathers.
- A bird with at least one dominant **A** allele and one dominant **D** allele has green feathers.
- A bird that is homozygous for **a** and **d** has white feathers.

Two green-feathered budgerigars, heterozygous at both gene loci, were crossed. Draw a genetic diagram of this cross to show the probability of producing offspring with yellow feathers.

[6]

[Total: 10]

7 Fig. 7.1 shows a neurone forming three synapses with adjacent neurones.

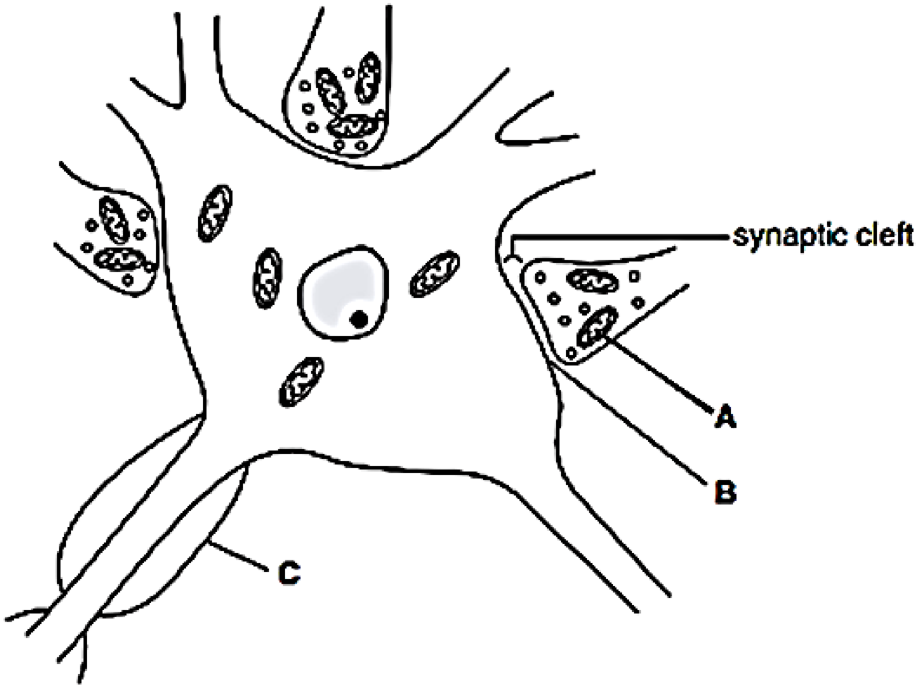

Fig. 7.1

(a) Name **A**, **B** and **C**. [3]

A ....

B ....

C ....

(b) State how synapses ensure one-way transmission of nerve impulses.

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....[1]

(c) In a learning activity, it is believed that the number of synapses between brain neurones increases.
Suggest the advantages of this increased number of synapses.

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....[2]

(d) Fig. 7.2 shows endings of neurones **X** and **Y** in the heart.

Stimulation of **Y** leads to an increase in the concentration of the second messenger, cyclic AMP (cAMP), in the cytoplasm of the heart cell and the stimulation of **X** leads to a decrease.

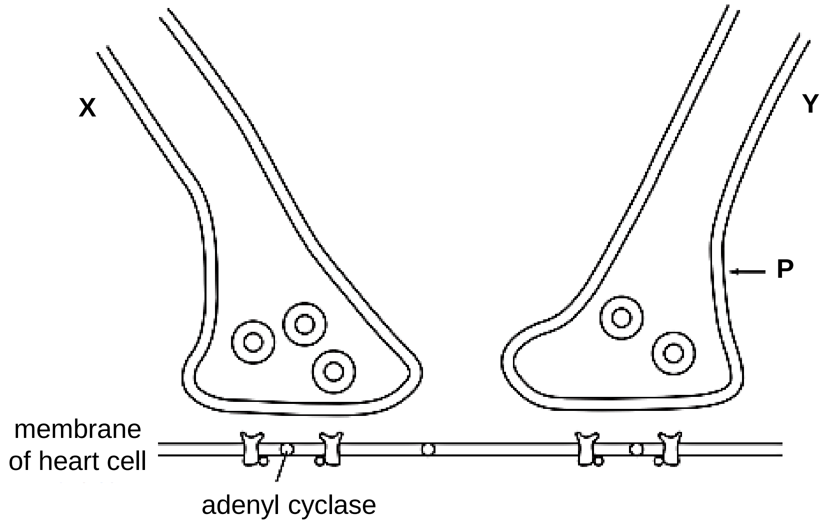

Fig. 7.2

Describe the events that follow the arrival of an impulse at **P** on neurone **Y** that leads to an increase in cAMP in the cytoplasm of the heart cell.

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[Total: 10]

8 (a) Fig. 8.1 show the location of two types of pistol shrimp with reference to the Panama strip.

Panama is a narrow strip of land which today joins North America and South America. It was formed by land moving up from beneath the sea. Panama has separated the Pacific Ocean and the Caribbean Sea for the past 3 million years.

Scientists put one **Type A** shrimp and one **Type B** shrimp together in a tank of seawater. The two types of shrimp snapped their claws aggressively at each other. They did not mate. The scientists said that this was evidence for the **Type A** and **Type B** shrimps being classified as two different species.

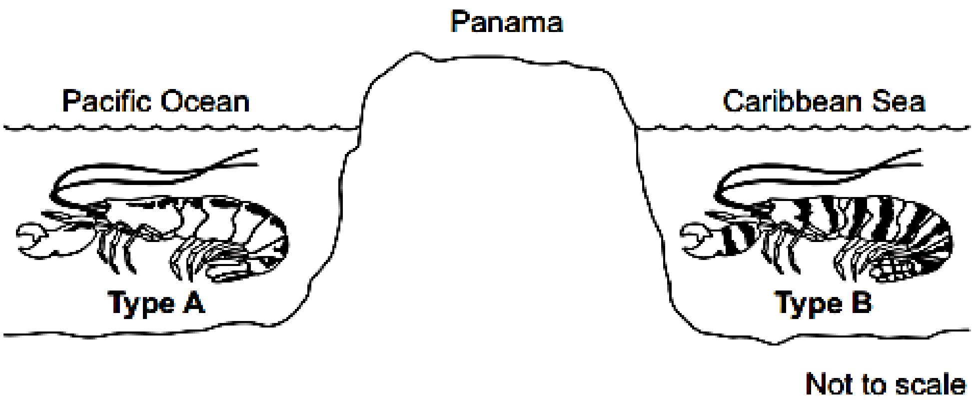

Fig. 8.1

With reference to the context given, explain how two different species of pistol shrimp could have developed from an ancestral species of shrimp.

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(b) Scientists used DNA hybridisation to determine the evolutionary relationships between five species of primate. The temperature at which a molecule of double-stranded DNA separates into two single strands is the separation temperature. The scientists recorded the mean separation temperature of DNA in which both strands were from the same species. The scientists then recorded the mean decrease in separation temperature of DNA in which one of the strands was from another species. Their results are shown in Table 8.

Table 8

Primate	**Mean decrease in separation temperature/ °C**				
	Human	Chimpanzee	Gorilla	Orang-utan	Gibbon
Human					
Chimpanzee	1.7				
Gorilla	2.3	2.3			
Orang-utan	3.6	3.6	3.5		
Gibbon	4.8	4.8	4.7	4.9	

(i) These data suggest that gibbons are the most distantly related to humans. Explain how.

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(ii) There were differences in separation temperature of DNA formed from single-stranded DNA of the **same** species of primate. Suggest why.

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(iii) The scientists assumed that the decreases in separation temperatures are directly proportional to the time since the evolutionary lines of these primates separated. Gorillas are thought to have separated from orang-utans 20 million years ago.

Use this information to calculate how long ago the evolutionary lines of humans and chimpanzees separated. Show your working. [2]

……………………… million years

[Total: 10]

[Turn over

Section B

Answer **one** question.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

9 **(a)** Describe the various types of genetic changes in proto-oncogenes which can result in cancer. [7]

(b) Explain, with named examples, how environmental factors act as forces of natural selection. [8]

(c) The life cycles of bacteriophages can be classified as ‘lytic’ or ‘lysogenic’. These terms are sometimes also used for animal viruses.

Use your knowledge of ‘lytic’ and ‘lysogenic’ to contrast the lytic life cycle of the influenza virus with the lysogenic life cycle of HIV. [5]

[Total: 20]

10 **(a)** Describe the pathway taken by cell-membrane bound proteins following their synthesis. [7]

(b) Describe the roles of NAD and NADP in cells. [8]

(c) Describe how the differences in the structure and organization of genetic material in eukaryotes and prokaryotes affects the way their genes are expressed. [5]

[Total: 20]